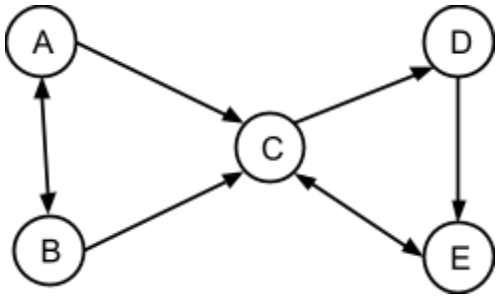
American Computer Science League

2025-2026 ● Contest 4: Short Problems ● Junior Division

1. Graph Theory

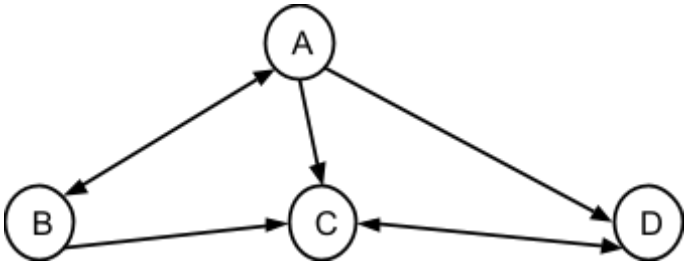
How many paths of length 2 are there in the following graph?



A. 6
B. 8
C. 9
D. 11
E. 13

2. Graph Theory

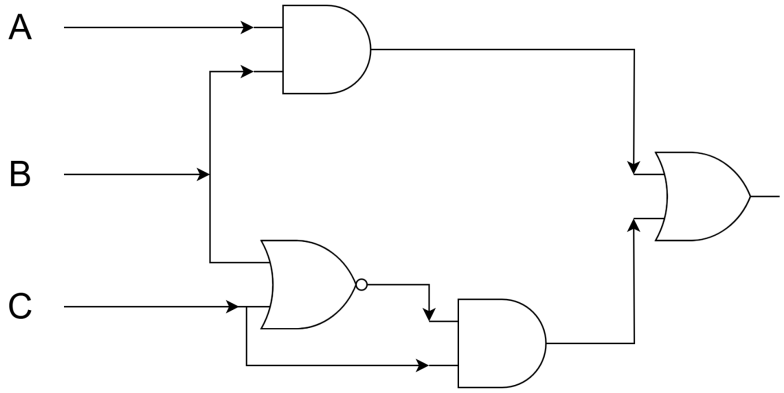
How many cycles are there in the directed graph below?



A. 0
B. 1
C. 2
D. 3
E. 4

3. Digital Electronics

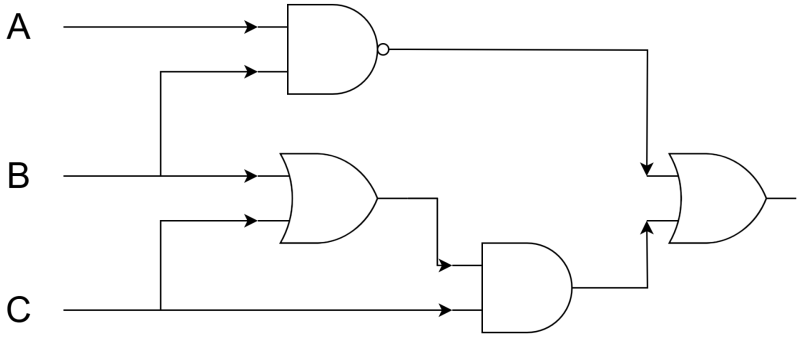
Simplify the Boolean algebra expression represented by the following digital circuit:



A. $\overline{A}B$
B. $A\overline{B}$
C. $A + B$
D. AB
E. $AB + \overline{B}C$

4. Digital Electronics

How many ordered triples make the following digital circuit TRUE?



A. 1
B. 3
C. 5
D. 7
E. 8

5. What Does This Program Do? (Strings)

What will be the output for the following program if the function val(*s*) converts the string *s* to its numerical value and str(*a*) converts the numeric value *a* to a string?

```
s = "742"
r = ""
for x = 0 to 2
     r = s[x] + r
next x
a = str(val(s) - val(r))
b = ""
for x = len(a)-1 to 0 step -1
     b = b + a[x]
next x
output val(b) + val(a)
```

A. 99
B. 189
C. 989
D. 1089
E. 495954

6. What Does This Program Do? (Strings)

What will be the output for the following program given that all uppercase letters come before all lowercase letters alphabetically?

```
s = "APCompSciAinJava"
ans = ""
for x = 0 to len(s) - 2 step 2
     if s[x] < s[x+1] then
          ans = ans + s[x] + s[x+1]
     else
          ans = ans + s[x+1] + s[x]
     end if
next x
output ans
```

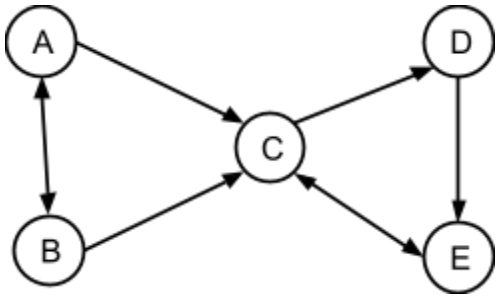
A. APCompSciAinJava
B. APCompScAiinJaav
C. APCOMPSCAiinJava
D. PACOMPSCAiinJava
E. PAoCpmcSiAniaJva

American Computer Science League

2025-2026 ● Contest 4: Short Problems ● Junior Division

1. 图论

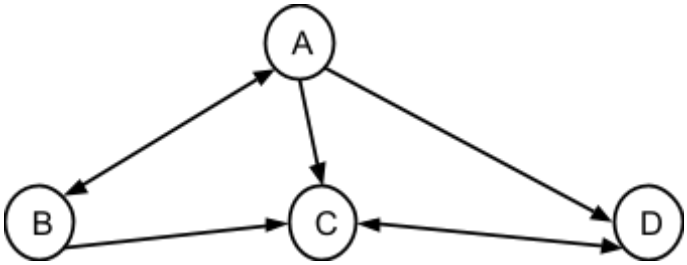
请问在下图中，长度为 2 的路径共有多少条？



A. 6
B. 8
C. 9
D. 11
E. 13

2. 图论

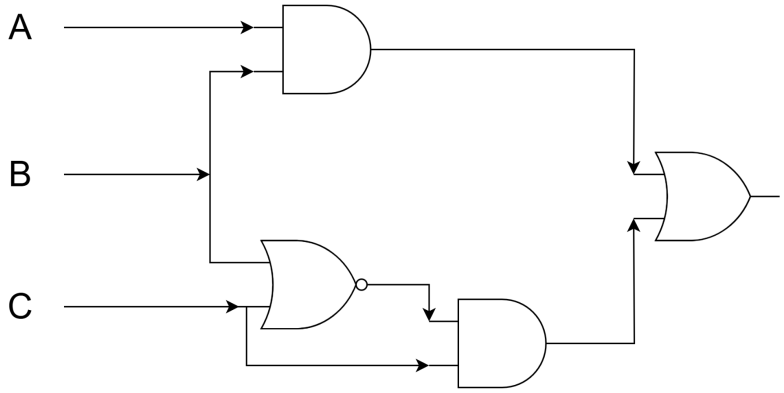
请问以下有向图中存在多少个环？



A. 0
B. 1
C. 2
D. 3
E. 4

3. 数字电路

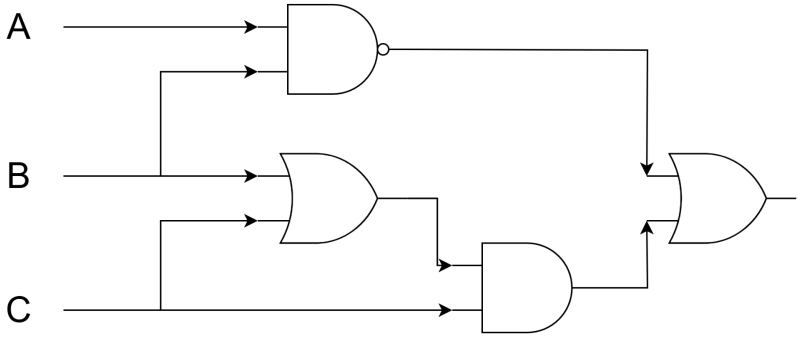
请化简下图所示数字电路所对应的布尔代数表达式：



A. $\overline{A}B$
B. $A\overline{B}$
C. $A + B$
D. AB
E. $AB + \overline{B}C$

4. 数字电路

请问有多少个有序三元组能够使得以下数字电路为真？



A. 1
B. 3
C. 5
D. 7
E. 8

5. 程序理解（字符串）

若函数 val(*s*) 用于将字符串 *s* 转换为对应数值，str(*a*) 用于将数值 *a* 转换为字符串，则下面程序的运行结果是什么？

```
s = "742"
r = ""
for x = 0 to 2
     r = s[x] + r
next x
a = str(val(s) - val(r))
b = ""
for x = len(a)-1 to 0 step -1
     b = b + a[x]
next x
output val(b) + val(a)
```

A. 99
B. 189
C. 989
D. 1089
E. 495954

6. 程序理解（字符串）

已知所有大写字母在字母表顺序中均排在所有小写字母之前，则以下程序的输出结果是什么？

```
s = "APCompSciAinJava"
ans = ""
for x = 0 to len(s) - 2 step 2
     if s[x] < s[x+1] then
          ans = ans + s[x] + s[x+1]
     else
          ans = ans + s[x+1] + s[x]
     end if
next x
output ans
```

A. APCompSciAinJava
B. APCompScAiinJaav
C. APCOMPSCAiinJava
D. PACOMPSCAiinJava
E. PAoCpmcSiAniaJva